



GE Medical Systems

Technical Publications

Direction 2329420
Revision 2

Signa LX Release 9.0 Patch 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 17 Software Load Instructions

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!!! IMPORTANT !!! This procedure requires additional steps to complete the software update. Just loading the CD will not provide all of the updates. Please see section 3.2 and section 5. **!!! IMPORTANT !!!**

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1- REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
0	1/23/2002	Initial version of procedure.
1	1/31/2002	Additional patches 11G, 12G, 13H, 5H, 2G, 8H, 6G
2	5/6/2002	Additional patches

PAGE	REV 0	PAGE	REV 1	PAGE	REV	PAGE	REV
Title Page	0	Title Page	1	Title Page.....	2		
5 to 32	0*	4 to 232.....	1*	4 to 32	2*		

* This revision/letter corresponds to the indicated document's revision control system.

2- OVERVIEW

The Signa LX Release 9.0 Patch 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 17 CD contains updates for the following software features / issues found in Release 9.0 M4.

#	Description
4	Non-applicable error message – HostVmSampleData.dat file could not be found
4	Coil Config Manager does not support ENDOATD and ATD3ENDO on 1.5T
4	Cannot Archive or network if service CD or IIP load / config done while Signa is up
4	Scans on wrong Gradient coil after amplifier trip (TRM only)
4	Utilities menu covers start button on service desktop
4	Gama lookup table lost after patch 4D
4	patch 4D disable or archive/network
4	iLinq gen2 lost after restore info
4	System crashes after noise only test
4	Unable to edit 1 st coil in site using CFM
4	Ucerd type not set correctly from Guided Install
4	Shutdown hangs
4	LV shim protocol has APS peak on water instead of peak
5	Images out of order while viewing a multi-acquisition scan in the browser
5	Reference images don't stay displayed during multi-acquisition scans
5	Cross Reference image jumps while viewing in the browser
5	Cannot archive 5.x images
5	Image Archiving and DICOM standard error
5	Images jump while viewing images with the mini-viewer
5	Spectro archive/restore translator error
5	Cross-reference image jumps in mini-viewer
6	Functool 2D CSI non-functional with release 9.0 without the 3D CSI key
6	Reformat fails with larger slice thicknesses than the default in "Batch" mode
7	Prescription / Download errors: "opmonthick is out of range," "ihoffsetfreq is out of range"
7	"ihoffsetfreq is out of range" caused by download error
8	Poor Image contrast with CinePhaseContrast
8	Double Obliques in Fiesta 2D
9	Elliptic centric ordering in efgre3D without a SmartPrep key
11	No Fat SAT with SSFSE when Manual Prescan is used
11	Fractional NEX problems in SSFSE
12	FSE with respiratory triggering
12	1RR/2RR slices difference in fse (fseopt and fse-xl)
13	EPI Ghosting after graffidy cals (TRM only)
13	Remove DOT artifact in DWEPI
14	Fiesta 2D not available without fiesta 3D option key
14	3 plane graphic RX may show localizers from prior exam
14	Smart prep tracker in the wrong place
14	Spectro oblique rotations are in the wrong place
14	Phase/freq direction issue with poor MRCP IQ
14	Film, archive, network resume after RTIA
14	PROSE prescan fix for multi coil
14	PROSE shifted lipid peak
14	GrxServer memory leak

- 14 Images upder different patient name
- 14 PROSE recon line broadening
- 15 Early bolus detection in SmartPrep
- 17 #DFSE aborts on second slab

Note

Signa software must be at the M4 version of Release 9.0 to install this patch CD. During the installation of the patches, the system must be idle and rebooted immediately after the installation.

3- REQUIRED PARTS**3-1 Additional parts**

- Patch CD 2327548-3 Rev 004 Signa Lx Patch for 9.0 M4
- Tech Pub 2329420 Rev 2 Signa Lx Release 9.0 Patch Software Load Instructions

3-2 DETERMINING PREVIOUS PATCH REVISION

IMPORTANT

THIS SECTION DETERMINES WHICH PATCHES HAVE BEEN PREVIOUSLY LOADED ONTO THIS SYSTEM. THIS STEP IS CRITICAL IN DETERMINING WHICH STEPS ARE NECESSARY FOR POST PROCESSING.

Prior to loading the Patch CD, determine which patches have already been loaded on your system. This will save time in the Configuring the System section (5) of this procedure.

- a) Inside a C Shell window, type the following
- b) **cd /usr/g/bin/ <enter>**
- c) **getver <enter>**

Build number for patch_90_M4_02 is 90.29F_M4_0139.z.PG
Build number for patch_90_M4_04 is 90.29F_M4_0139.z.PD
Build number for patch_90_M4_05 is 90.29F_M4_0139.z.PH
Build number for patch_90_M4_06 is 90.29F_M4_0139.z.PG
Build number for patch_90_M4_07 is 90.29F_M4_0139.z.PD
Build number for patch_90_M4_08 is 90.29F_M4_0139.z.PH
Build number for patch_90_M4_09 is 90.29F_M4_0139.z.PD
Build number for patch_90_M4_11 is 90.29F_M4_0139.z.PG
Build number for patch_90_M4_12 is 90.29F_M4_0139.z.PG
Build number for patch_90_M4_13 is 90.29F_M4_0139.z.PH

If there are no installed patches, complete all of section 5.

If patch 6 is at revision **patch_90_M4_06 is 90.29F_M4_0139.z.PG** Section 5-1 **does not** need to be completed.

If patch 13 is at revision **patch_90_M4_13 is 90.29F_M4_0139.z.PH** Section 5-4 **does not** need to be completed.

4- LOADING SIGNA LX RELEASE PATCH FOR 9.0 M4 FROM CD-ROM

1. Make sure the system is idle (not scanning, filming, viewing or networking). The best way to confirm the system is idle is to restart the system and log in as root (in this case, proceed to step 3).
2. Inside a C Shell window, log in as the root user by typing the following:

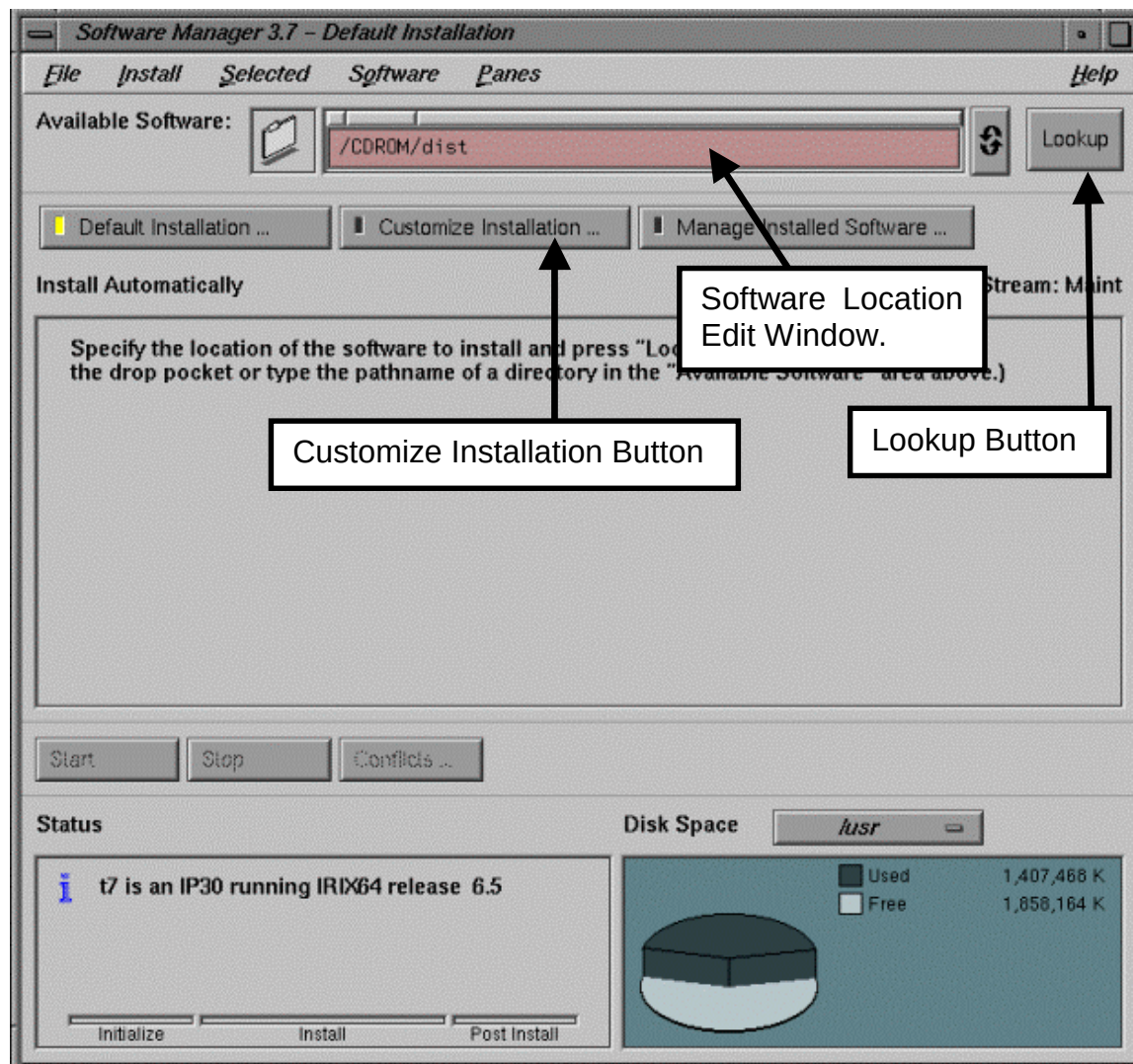
su - <Enter>

Password: **operator <Enter>** ("operator" is the system default root password. Use whatever password is appropriate for the system you are working on if it has been changed).

Note

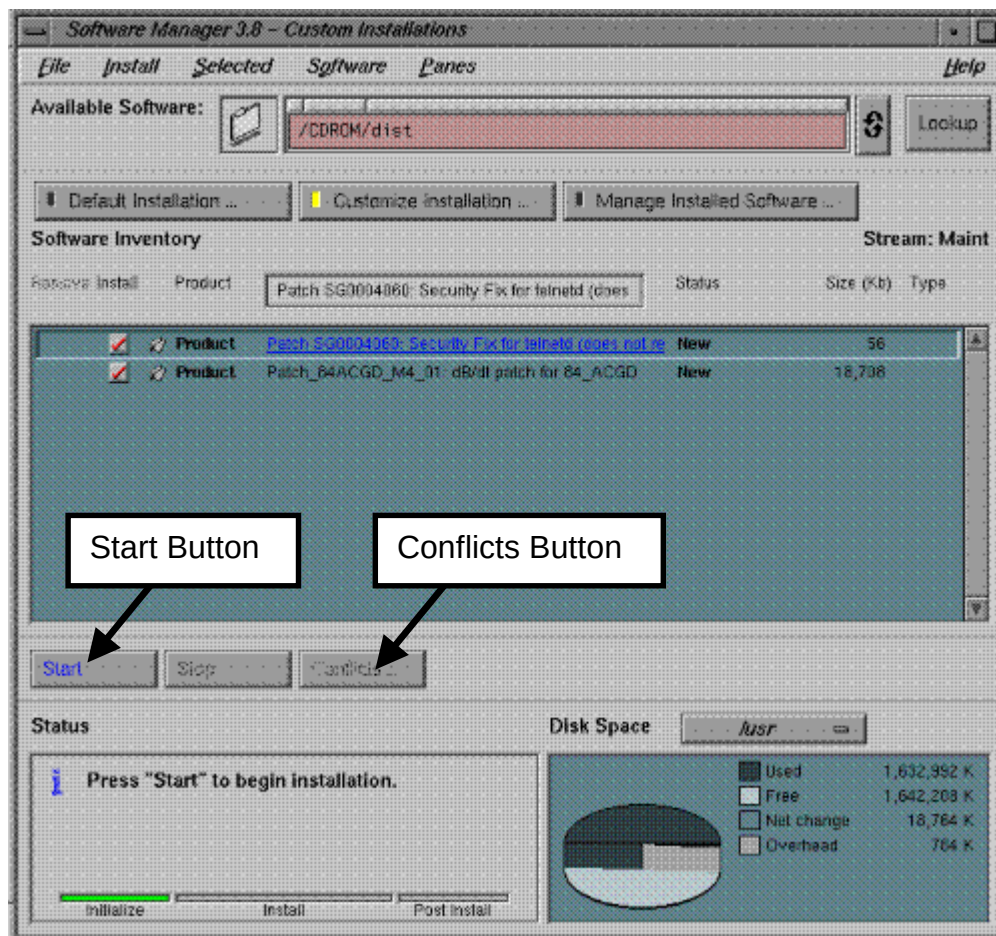
If the patch has already been downloaded from the AutoSC or the OLC, proceed to step 4b.

3. Place the "Signa 9.0 M4 Patch" 232754-3 CD in the system (not PC) CD drive. Wait 20 seconds for the system to see the CD until the light on the front of the CD drive stops blinking.
4. Inside the C Shell, type the following:
 - a) **mediad <Enter>** (This command mounts the CDRom).
 - b) **swmgr <Enter>** (This command starts the software manager which is used to load the patch).



SOFTWARE MANAGER INITIALIZATION
ILLUSTRATION 4-1

- Once the Software Manager starts, begin typing the directory location where the patches are stored (case sensitive) in the **[Software Location Edit Window]**. See Illustration 4-1. If the patches were downloaded from the OLC or the AutoSC proceed to step 5B.
 - /CDROM** (The system will automatically enter **"ldist"** after you type **/CDROM**)
- or -
 - /usr/g/insite/data** (If the patches were downloaded from the AutoSC or the OLC)
- Click the **[Lookup]** button found to the right of the **[Software Location Edit Window]**. See Illustration 4-1.
- Click the **[Customize Installation]** button at the top of the screen. See Illustration 4-1.



CUSTOMIZE INSTALLATION SCREEN EXAMPLE
ILLUSTRATION 4-2

8. All available patches should be visible at this time in the selection window. Any patches that apply to your system and are not loaded will be selected (check marked). Verify the appropriate box is checked corresponding the system software version. Note: only the correct software version should be checked. See Illustration 4-2. If the second patch CD was installed, the checked patches will be identified as in the following list.

✓	Product	Patch_90_M4_04 Tools Patch J for 90 M4 Release	Patch Upgrade
✓	Product	Patch_90_M4_07 Protocols patch L for 90 M4 Release	Patch Upgrade
✓	Product	Patch_90_M4_012 FSE Patch K for 90 M4 Release	Patch Upgrade
✓	Product	Patch_90_M4_013 EPI Patch J for 90 M4 Release	Patch Upgrade
✓	Product	Patch_90_M4_014 Scan/Spectro/HOShim for 90 M4 Release	New
✓	Product	Patch_90_M4_015 Smart prep for 90 M4 Release	New
✓	Product	Patch_90_M4_017 3dfse for 90 M4 Release	New

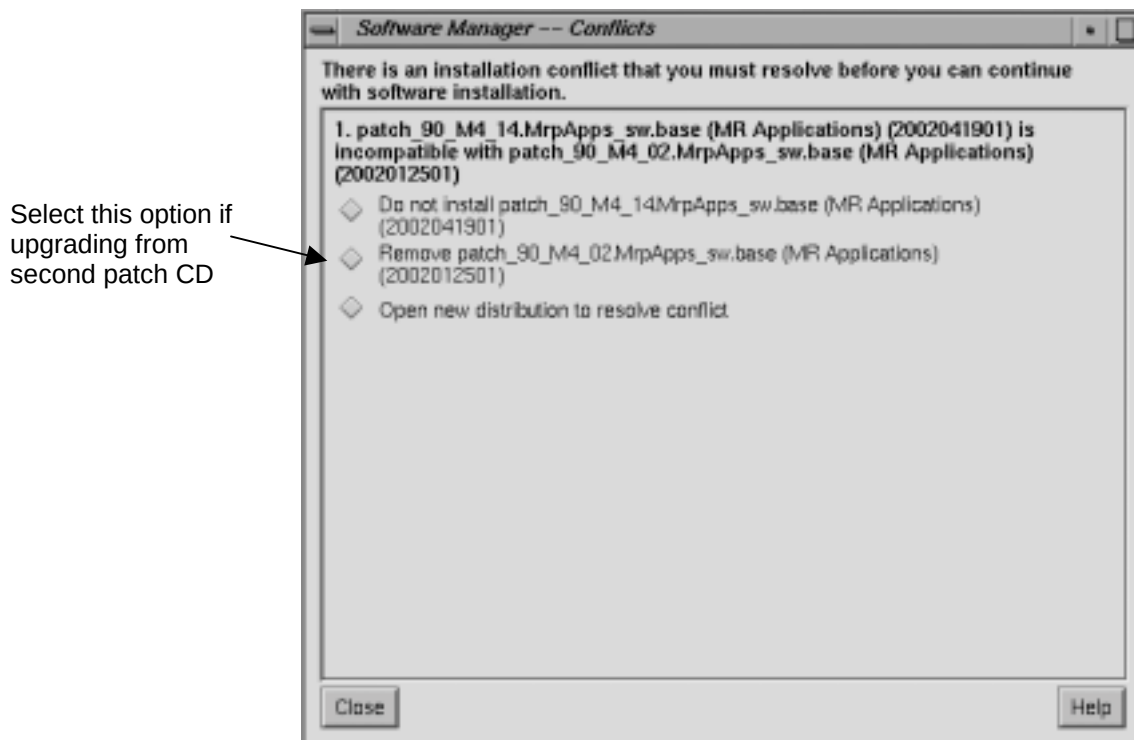
9. If the **[Start]** button activates, proceed to 9.a. If the **[Conflicts]** button activates proceed to section 9.b

Note

If the second patch CD was previously loaded the **[Conflicts]** will be activated. See step B for details.

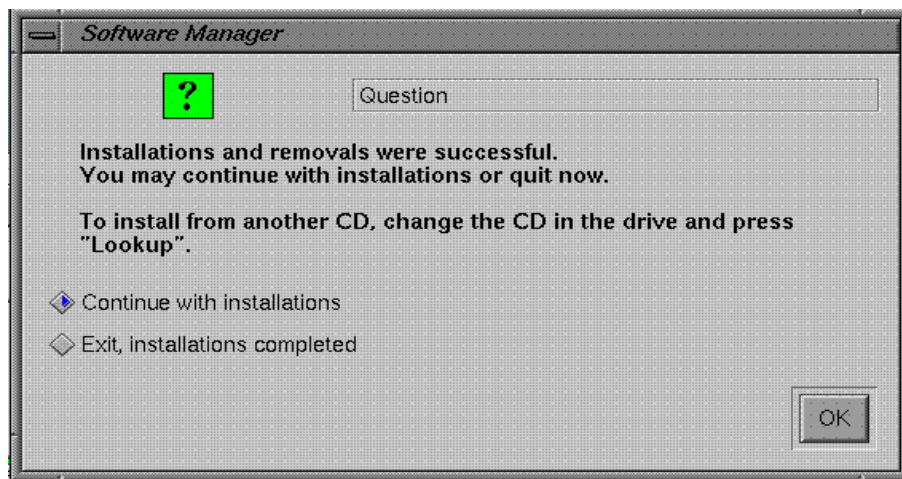
A) The **[Start]** button will activate (The letters in "Start" will change colors or shades). If the **[Start]** button activates, push it and the patch load will begin. A progress bar will open and count up from 0 to 100%. The load takes less than 1 minute. When the progress bar goes away, a Question window will be displayed. Select **[Exit, Installations completed]** and **[Ok]** to quit the Software manager. Proceed to step 10.

B) If site is upgrading from Second Patch CD the **[Conflicts]** button will activate, (the letters in "Conflicts" change colors or shades). A conflict with patch 2 is expected. Select the option to **Remove patch_90_M4_02_MrpApps**. See Illustration 4-2b. If site had loaded patch 10 for High Order Shim option, this patch will also conflict. A conflict with patch 1, 3 or 10 is also possible, depending on what patches were previously loaded. These patches can be deleted. If there are **conflicts** with other patches , remove applicable patch and proceed with patch installation. Hit **start** to install patches.



CONFLICTS WINDOW
ILLUSTRATION 4-2B

10. After installation, select **Exit, Installations completed** checkbox, and select **[OK]**. See Illustration 4-3.



INSTALLATION DECISION WINDOW
ILLUSTRATION 4-3

11. In the C Shell type the following (case sensitive) to eject the CD-ROM from the drive (if the patches were downloaded from the AutoSC or the OLC, proceed to step 12):
umount /CDROM <Enter> (Un-mounts the CD from the system)
eject /CDROM <Enter> (Opens the CD drawer)

12. If the patches were required and loaded, verify they are installed by typing in the C Shell:
/usr/g/bin/getver <Enter> You should see an entry for the LX patch as follows
(for example only: the bold lines are related directly to this update):

Build number for MrpApps is 90.29F_M4_0139.z
Build number for PostSdC is 90.29F_M4_0139.z
Build number for cclass is 90.29F_M4_0139.z
Build number for driverSupport is 90.29F_M4_0139.z
Build number for insite is 2.3-mips
Build number for install is 90.29F_M4_0139.z
Build number for os_cd is 6_11.1.a
Build number for os_u_cd is 6u11.6.a
Build number for patch_90_M4_04 is 90.29F_M4_0139.z.PJ
Build number for patch_90_M4_05 is 90.29F_M4_0139.z.PH
Build number for patch_90_M4_06 is 90.29F_M4_0139.z.PG
Build number for patch_90_M4_07 is 90.29F_M4_0139.z.PL
Build number for patch_90_M4_08 is 90.29F_M4_0139.z.PH
Build number for patch_90_M4_09 is 90.29F_M4_0139.z.PD
Build number for patch_90_M4_11 is 90.29F_M4_0139.z.PG
Build number for patch_90_M4_12 is 90.29F_M4_0139.z.PK
Build number for patch_90_M4_13 is 90.29F_M4_0139.z.PJ
Build number for patch_90_M4_14 is 90.29F_M4_0139.z.PL
Build number for patch_90_M4_15 is 90.29F_M4_0139.z.PJ
Build number for patch_90_M4_17 is 90.29F_M4_0139.z.PJ

Build number for pc is 90.29F_M4_0139.z
Build number for z is Upgrade
MR Software release: 9.0.0139z

Note:

If the patches were downloaded from the OLC via an online engineer or the AutoSC, they must be deleted after the installation completes to free up user disk space. They are located in /usr/g/insite/data.

5- CONFIGURE THE SYSTEM TO RUN WITH THE UPDATE

5-1 Enable the Functool option if the site has the MOD

IMPORTANT

This step will not be performed if the previous revision of patch 6 was **patch_90_M4_06 is 90.29F_M4_0139.z.PG**. Refer to section 3-2.

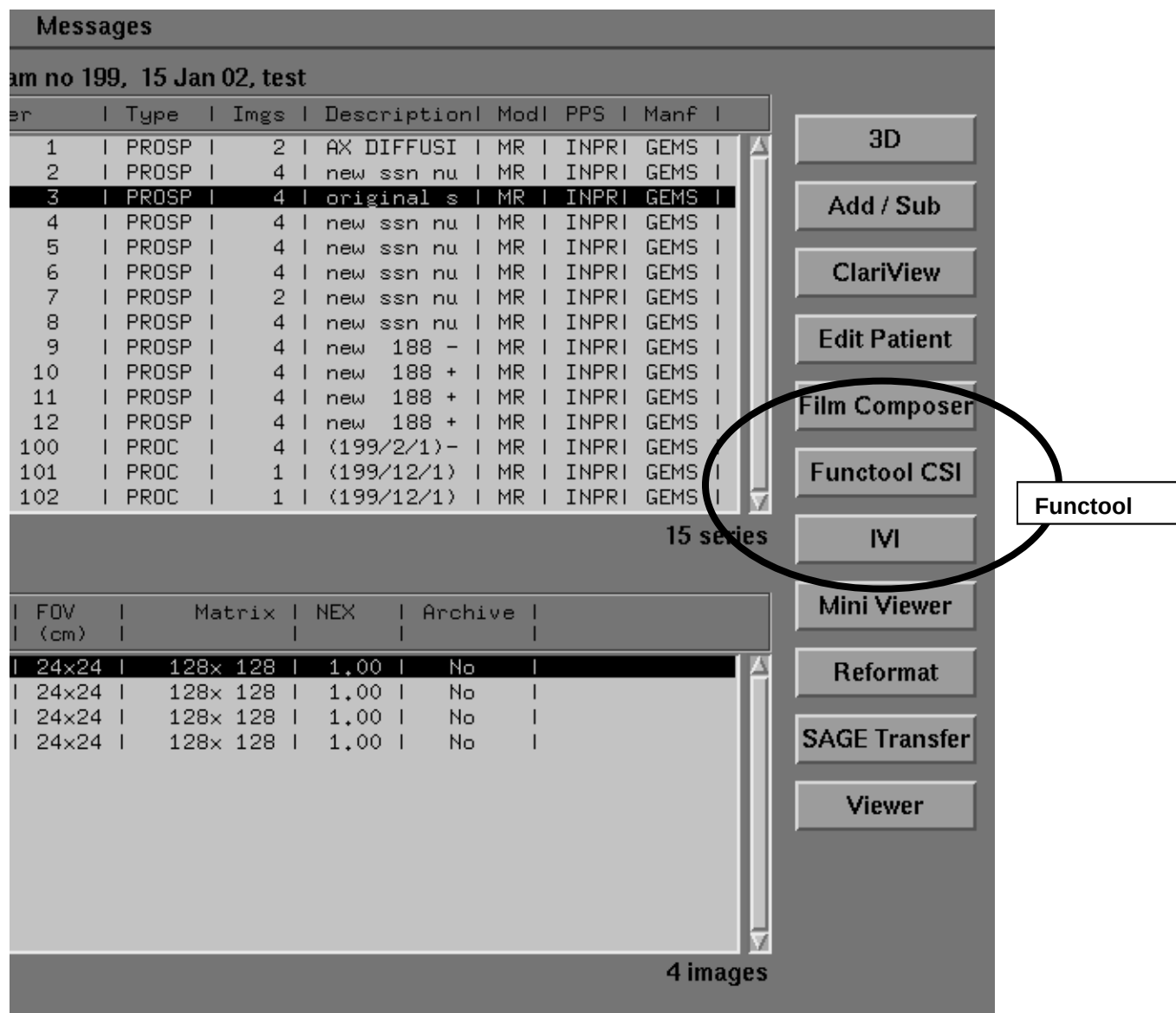
- 1) If the site does not have the Functool option MOD(s), proceed to updating the system protocols.
- 2) Launch the **Guided Install** utility in FE mode.
- 3) Select the **Option Keys** tab
- 4) Uninstall any Functool options (Functool and Functool CSI) if they show up as permanent or eFlextrial in the option GUI. Select the option and click on **Uninstall**. Uninstall the Functool CSI option first and then uninstall the Functool option.
- 5) Insert the Functool MOD into the MOD drive.
- 6) Select **MOD** and select **Install**
- 7) When the message box comes up, insert the next Functool MOD (if applicable) and wait fifteen seconds.
- 8) When the Message box comes up, Select **Quit**.
- 9) Using the option key display, verify that the appropriate Functool key(s) are listed as permanent.

Select	Key Name	Key Type	Expiration Date
☐	SmartPrep 2000 Upggrade	permanent	0
☐	Probe 2000 Upgrade	permanent	0
☐	Multi-Nuclear Spectroscopy	permanent	0
☐	Functool	Permanent	-----
☐	Functool CSI	Not Installed	-----
☐	Voxtool	Not Installed	-----
☐	Intervascular Imaging	Not Installed	-----

OPTION KEYS INSTALLED

ILLUSTRATION 5-1

- Complete the next section and reboot the system, then, verify the option key has been installed correctly by looking for the “Functool” or “Functool CSI” button on the browser menu on the browser / viewer desktop after the system has been booted as Signa. If both buttons show up, deinstall the options and then reinstall. See illustration 5-1.



BROWSER MENU WITH FUNCTOOL CSI BUTTON
ILLUSTRATION 5-2

- Exit **Guided Install**.

5-2 Update all of the system protocols (GE and site)

1. Inside a C Shell window, log in as the root user by typing the following:

su - <Enter>

Password: **operator <Enter>** ("operator" is the system default root password. Use whatever password is appropriate for the system you are working on if it has been changed.)

2. Enter in "**cd /usr/g/bin**"
3. Enter in "**ProtocolCorrect.sh**"

Note:

This will edit all protocols under /usr/g/protocols, including site, ge, etc. Before this editing happens a backup of the entire /usr/g/protocols directory will be tarred and gzipped and saved as /usr/g/protocols.bak.tar.gz.

All of the protocols from the usr/g/protocols directory will scroll by. This process takes 5-7 minutes depending on the number of site-specific protocols on the system.

4. Reboot the system and start Signa
5. If the Functool CSI key was loaded in the previous section, select the browser desktop and verify the Functool CSI button exists on the browser menu. See illustration 5-1. If the button is not correct, uninstall and reinstall the Functool CSI key and reboot.
6. Run two test scans. One with a GE protocol and one with a site-specific protocol.
7. If a critical error occurs contact the OLC for support and reference the clarify solution database "Release 9.0 ProtocolCorrect.sh script"

5-3 Check CERD / UCERD Setting in MRConfig

Note:

The **Guided Install** Tool does not edit the CERD type field in MRConfig. Use **Config File Editor** to change the setting or severe image quality problems may be seen in SmartStep or SmartPrep.

1. Start **Config File Editor** from the service desktop.
2. Select the **CERD** type config field.
3. Verify the correct CERD type is set and change if necessary. Select **001** for CERD or **002** for UCERD. Do not use 021 or 022 at this time.

5-4 TwinSpeed only: Reset Super Secret Numbers (2.5 hours)

IMPORTANT

This step will not be performed if the previous revision of patch 13 was **patch_90_M4_13 is 90.29F_M4_0139.z.PH**. Refer to section 3-2.

Note:

To take advantage of the ghosting reduction on TwinSpeed systems, you must load the new eddy current coefficients into the eddy current compensation files by starting (not running) graffidy and running head and body EPT calibrations (not white pixel).

4. Start the **Graffidy** tool from the service desktop.
5. Select **Whole** body mode and **Ok**.
6. Do not run graffidy. Exit the tool by typing **Q <Enter> <Enter>**
7. Repeat for Zoom mode. Start the **Graffidy** tool from the service desktop.
8. Select **Zoom** coil mode and **Ok**.
9. Do not run graffidy. Exit the tool by typing **Q <Enter> <Enter>**

Note:

For detailed instructions on EPT, refer to Tech Pub
SYSSCC3.doc ECHO PLANAR TEST (EPT)

10. Start the **EPT** tool from the service desktop.
11. Select **Head: B Zero Dither, B Zero Image**.
12. Select **Body: B Zero Dither, Group Delay, B Zero Image**.
13. Select **Start**

5-5 Perform a SaveInfo

1. Launch the **Guided Install** utility in FE mode and insert the site's SaveInfo MOD into the drive.
2. Use the SaveInfo tab to update the site's SaveInfo MOD with the recent changes.

****IMPORTANT****

If IVI option is removed and re-installed, **patch 6 must** be re-installed again.

6 – APPENDIX DETAILED PATCH DESCRIPTIONS

Note: These descriptions are excerpted from the readme.txt files that accompany each patch.

Patch_90_M4_04 Rev J

PROBLEM 1 - Unable to open HostVmSampleData.dat message
partially addressed in version D of this patch

SPR - MRlge69010 Hackensack: Odd message ULHC in message box, won't go away
SPR - MRlge68800 Erronous mesg. Keeps appearing, but doesn't seem to cause prob
SPR - MRlge73895 "Unable to open HostVmSampleData.dat" error message on 90, 91

Description -

The vacuum monitor is defaulting to active on non-TRM systems. The error message is a nuisance only and does not cause any system malfunction. The resolution is to change the system start-up script, so as to start only on TRM.

Workaround -
None.

The fix is in the following subsystems:
HOST: startmon script

PROBLEM 2 - Coil Config Manager does not work for ENDOATD and ATD3ENDO on 1.5T
previously addressed in version D of this patch

SPR - MRlge71202 Errors in CoilConfig files prevent use of some coils in 9.0

Description -

The original 1.5T Coil Config files for Release 9.0 did not include information for two prostrate spectroscopy coils: ENDOATD and ATD3ENDO. The patch includes coil config files for the two coils.

Workaround -
None.

The fix is in the following subsystems:
HOST: Coil Config files

PROBLEM 3 - Cannot Archive or Network if Service or IIP Load done with Signa up
previously addressed in version D of this patch

SPR - MRlge69757 insite/iip changed ownership on critical files

Description -

Unable to network images or archive images after a Restricted Service Load or IIP Load is done with Signa running. This is because critical files in sdc account had ownership changed to insite/iip, preventing user account sdc from accessing them as needed.

Workaround -

Remove the soft link between the service directory and /tmp by typing:

```
rm /usr/g/csd/mr_csd/serviceDesktop/tmp
```

Then make sure the permissions on the /tmp directory are "drwxrwxrwt" and the owner is set to "sys".

The fix is in the following subsystems:

HOST: remove the soft link between /usr/g/csd/mr_csd/serviceDesktop/tmp and /tmp

PROBLEM 4 - Utilities menu covers start button on service desktop
previously addressed in version D of this patch

SPR - MRIge71003 Utilities menu covers start button on service desktop

Description -

In the Service Desktop Manager (SDM), when one selects the Utilities Button, the horizontal scroll bar overlaps the START Button which is just below the Utilities Menu. To fix this problem, the height of the scrolled Window was reduced.

Workaround -

Use the edge of the menu to select the desired choice.

The fix is in the following subsystems:

HOST: SDM

PROBLEM 5 - Scans on wrong grad coil after amp trip
previously addressed in version D of this patch

SPR - MRIge65565 System scanning with wrong grad coil after amp trip

SPR - MRIge70699 Edison - FOV size is not what is prescribed

SPR - MRIge70001 System scanning with wrong grad coil after amp trip

Description -

Intermittently, for Twin systems only, after an amplifier trip, the scanner was scanning with the wrong gradient coil, resulting in images that appeared as if they were scanned at the wrong FOV. Additionally, sometimes MDS link errors 2225693, 2225699, and 2225704 are logged.

Workaround -

Reboot Signa.

The fix is in the following subsystems:

Realtime: SPI

PROBLEM 6 - gamma lookup table lost after patch 4D

CQA - 1021834 9.0 Monitor, rungamma

SPR - MRIge72689 9.0 Monitor, rungamma

CQA - 1021027 9.0 LCD config, rungamma

SPR - MRIge72255 9.0 LCD config, rungamma

CQA - 1021111 9.0 Gamma tbl, rungamma

SPR - MRIge72321 9.0 Gamma tbl, rungamma

SPR - MRIge73227 startmon script corrupted - missing endif

The fix is in the following subsystems:

HOST: startmon script

PROBLEM 7 - patch 4D disable of archive/network

SPR - MRIge72918 Automation to enable archive / networking

The fix is in the following subsystems:

PATCH: installation script

PROBLEM 8 - ilinq gen2 lost after restore info

SPR - MRlge72881 ILinq Gen2 needs save/restore patch

The fix is in the following subsystems:

HOST: save/restore scripts

PROBLEM 9 - system crashes running Noise Only Test

CQA - 10110845 ASPII-System crashes when running Noise Only Test

SPR - MRlge70673 ASPII-System crashes when running Noise Only Test

The fix is in the following subsystems:

HOST: noise_script

PROBLEM 10 - unable to edit 1st coil in site list using CFM

CQA - 1020025 LEO1 Coilconfig name 4SHOULDR

SPR - MRlge71834 LEO1 Coilconfig name 4SHOULDR

SPR - MRlge71974 Unable to Edit 1st coil in Site list using ConfigFile Manager

The fix is in the following subsystems:

HOST: Config File Editor

PROBLEM 11 - ucerd type not set correctly from Guided Install

SPR - MRlge72567 hardware_write script needs to be modified

The fix is in the following subsystems:

HOST: Guided Install scripts

PROBLEM 12 - shutdown hangs

CQA - 1022320 9.0 shutdown hangs

SPR - MRlge73081 9.0 shutdown hangs

CQA - 1020064 LEO1 Shutdown hangs

SPR - MRlge71851 LEO1 Shutdown hangs

CQA - 1020398 9.0 no shutdown from tool menu

SPR - MRlge72050 9.0 no shutdown from tool menu

CQA - 1022213 9.0 shutdown not complete[D

SPR - MRlge73002 9.0 shutdown not complete[D

CQA - 1022606 9.0 shutdown hangs

SPR - MRlge73310 9.0 shutdown hangs

The fix is in the following subsystems:

HOST: stopmon script

PROBLEM 13 - LV shim protocol has APS peak on water instead of peak

SPR - MRlge73790 LV shim protocol has APS peak on water instead of peak

Description -

The first scan protocol under LVshim is a localizer and it has peak selected for APS. The second series is the one that you acutally scan and run the analysis on. This has APS peak on water. This is wrong and should be changed to PEAK.

Workaround -

Manually edit the protocol.

The fix is in the following subsystems:

Protocol: LVshim

Patch_90_M4_05 Rev H

This patch contains fixes for seven problems found with the 9.0 'LEO1' release, five of which were previously addressed in prior versions of this same patch (versions D, E, and G). The size of the download for this patch is approximately 2.6 megabytes, compressed.

The patch may be installed on a LFC Release 9.0 system, or over the top of versions D, E, or G of patch_90_M4_05 by the same procedure.

PROBLEM 1 - Images out of order during filming and networking
previously addressed in version D, E, and G of patch_90_M4_05

CQA - 10111854 LEO1 multiple issues
SPR - MRIge71291 LEO1 multiple issues
SPR - MRIge70564 West Allis: images not ordered correctly in viewer

Description -

While scanning a 3 acquisition interleaved series, if the viewer was opened after only the first acquisition was finished, the image order was 1,4,7. After the second acquisition, the image order was 1,4,7,2,5,8. After the third acquisition, the image order was 1,4,7,2,5,8,3,6,9.

Workaround -

Images could be put in the correct order by going back to the browser, and re-opening the series.

The fix is in the following subsystems:

HOST: Viewer

PROBLEM 2 - Reference Images Won't stay displayed during acquisition
previously addressed in version D, E, and G of patch_90_M4_05

CQA - 10111854 LEO1 multiple issues
SPR - MRIge71291 LEO1 multiple issues
SPR - MRIge70564 West Allis: images not ordered correctly in viewer

Description -

While viewing images on the second or greater pages with a multi-acquisition scan in progress the viewer intermittently jumps back to the first page.

Workaround -

Images could be put in the correct order by going back to the browser, and re-opening the series after the acquisition has completed.

The fix is in the following subsystems:

HOST: Viewer

PROBLEM 3 - Cross Reference lines lost during acquisition
previously addressed in version E and G of patch_90_M4_05

SPR - MRIge70982 St. Lukes: Page Down button reverts to image 1, 30% of time
SPR - MRIge70403 Pensacola-XR lines not screen saving
SPR - MRIge69895 Viewer: When #imgs in ri series changes; ri go away

Description -

During an exam, when the viewer is on a scanned series and there is scanning, images on the scanned series jump when new images of the scanning series arrive. This also happens when the viewer is switched to the currently scanning series and back to the scanned series. At this time, if cross-referencing is being done on the scanned series, the image jumping causes the cross-reference lines to vanish from the intended image.

For example: Here is the order of images on a 4x1 format

4 5

8 9

If you do a cross-reference on image 8, when image 6 or 7 gets installed, image 8 will lose its cross-reference.

Workaround -

The fix is in the following subsystems:

HOST: Viewer

PROBLEM 4 - Cannot archive 5.x images
previously addressed in version E and G of patch_90_M4_05

CQA - 10112699 LEO 5x archive on 9.0

SPR - MRIge71653 LEO 5x archive on 9.0

Description -

After sending a set of 5.x ("Genesis") exams to Release 9.0 system via either Advantage Net or DICOM, archive and restore will put all series of all the sent exams into one exam.

Workaround -

Send the exams one at a time.

The fix is in the following subsystems:

HOST: Archive, Translator

PROBLEM 5 - Cross Reference image jumps, not during acquisition
previously addressed in version G of patch_90_M4_05

SPR - MRIge70193 Pensacola-Cross Reference jumping images

Description -

The viewer jumps to the wrong image during Cross Reference, and there are no new images in acquisition at the time. For example:

- 1) To get Series 2, Image 1, type at command line: 2 1
- 2) To get Image 7, type at command line: 7
- 3) To cross reference Series 1 on this Image 7, type: xr 1
now the viewer jumps to Image 1 !
- 4) To see the cross reference, have to PAGEDOWN or again type: 7

Workaround -

Do not start by using command line to get starting series and image.

Use PAGEDOWN, PAGEUP, +, or - buttons. After that, use normal procedure.

The fix is in the following subsystems:

HOST: Viewer

PROBLEM 6 Spectro archive/restore translator error

SPR - MRIge71585 CQA 10112544 3T Spectrum, ratios changed

Description -

The problem is related to Spectroscopy utilities. When restoring Spectro image from an MOD, the data changes. The MI/CR field is incorrectly set to be equal to the CHO/CR ratio.

Workaround -

None.

The fix is in the following subsystems:

HOST: Translator

PROBLEM 7 Cross-Reference image jumps in miniviewer

SPR - MRIge72259 CQA 1021022 9.0 Miniviewer image changes

Description -

The miniviewer jumps to the wrong image during Cross Reference, and there are no new images in acquisition at the time. For example:

- 1) To get Series 2, Image 1, type at command line: 2 1
- 2) To get Image 7, type at command line: 7
- 3) To cross reference Series 1 on this Image 7, type: xr 1
now the miniviewer jumps to Image 1 !
- 4) To see the cross reference, have to PAGEDOWN or again type: 7

Workaround -

Do not start by using command line to get starting series and image. Use PAGEDOWN, PAGEUP, +, or - buttons. After that, use normal procedure.

The fix is in the following subsystems:

HOST: Viewer

Patch_90_M4_06 Rev G

This patch contains fixes for two problems found with the 90 'LEO1' release, none of which have been previously addressed. The size of this download for this patch is approximately 2.8 megabytes, compressed.

PROBLEM 1 - loss of functool 2D-CSI functionality

SPR - MRIge71269 Functool CSI eFlextrial may not resolve problems with 2D Func

CQA - 10112116 LEO1 functool, display overlay

SPR - MRIge71486 functool, display overlay

CQA - 10111854 LEO1 multiple issues

SPR - MRIge71291 LEO1 multiple issues

Description -

Release 9.0 Functool did not allow for 2D CSI features without the 9.0 Functool CSI option key installed. This patch updates the installation scripts for the Functool feature. The FE must then re-load the Functool CSI option key from the MOD.

Workaround -

Download an eFlextrial version of the 9.0 version of Functool CSI

The fix is in the following subsystems:

HOST: functool installation scripts - The Functool CSI option key must then be re-installed.

PROBLEM 2 - reformat failure in case of larger slice thickness than default

CQA - 10112054 LEO MIP issue, function loss

SPR - MRIge71390 LEO MIP issue, function loss

CQA - 10112253 LEO1 no MIP reformat

SPR - MRIge71476 LEO1 no MIP reformat

Description -

In reformat, if slice thickness larger than the prescribed thickness is entered, selection of any option rather than average (such as MIP), will cause reformat to crash.

Workaround -

None.

The fix is in the following subsystems:

HOST: voxtool

Patch_90_M4_07 Rev L

This patch contains fixes for two problems found with the 9.0 release, none of which have been previously addressed. The size of this download for this patch is approximately 5 kilobytes, compressed.

PROBLEM 1 - "opmonthick is out of range" nuisance message during Rx

CQA - 10111645 Leo1-opmonthick is out of range message

SPR - MRIge71149 Leo1-opmonthick is out of range message

Description -

The customer routinely sees a popup message "opmonthick is out of range" when prescribing scans with certain protocols. The problem is caused by protocols with "set TRACKTHICK = "0.00"" when it should be "set TRACKTHICK = ".

This patch downloads a script, which can be run to update all of the protocols in a particular directory. The patch does not automatically run this script after installation. The script should be manually executed. As root, type "cd /usr/g/bin; ProtocolCorrect.sh".

Workaround –

Customer can rebuild the protocol to correct the problem.

The fix is in the following subsystems:

HOST: protocol conversion script provided, but not run automatically.

PROBLEM 2 - "ihoffsetfreq is out of range" caused download error

Description -

The customer sees a download error message "ihoffsetfreq is out of range" routinely, with certain protocols. This problem is caused by protocols with USERCV0 through USERCV24 set to "0.00" when they should be set to "".

With GRE sequences saved with software other than 9.0, a message was displayed "ihoffsetfreq is out of range."

This patch downloads a script, which can be run to update all of the protocols in a particular directory. The patch does not automatically run this script after installation. The script should be manually executed. As root, type "cd /usr/g/bin; ProtocolCorrect.sh".

Workaround –

Customer can rebuild the protocol to correct the problem.

The fix is in the following subsystems:

HOST: protocol conversion script provided, but not run automatically.

PROBLEM 3 - "opmonthick" is out of range" nuisance error for all protocols

CQA - 1023412 9.0 Optmonthick out of range error

SPR - MRIge73822 9.0 Optmonthick out of range error

CQA - 1024092 9.0 Opmonthick is out of range

SPR - MRIge74321 9.0 Opmonthick is out of range

Description -

The earlier version of these scripts that modify protocols to solve this problem worked only of protocol files of filetype "ascii text", while sometimes protocols are filetype "command text". This updated script is run the same way, and can be run even if the earlier version has already been run.

The patch does not automatically run this script after intallation.

The script should be manually executed. As root, type:

```
cd /usr/g/bin  
ProtocolCorrect.sh
```

Workaround -

Customer can rebuild the protocol to correct the problem.

The fix is in the following subsystems:

HOST: protocol conversion script provided, but not run automatically.

Patch_90_M4_08 Rev H

This patch contains fixes for two problems found with the 9.0 'LEO1' release, one of which was previously addressed in a prior version of this same patch (versions D). The size of the download for this patch is approximately 2.4 megabytes, compressed. The patch may be installed on a LFC Release 9.0 system, or over the top of version D of patch_90_M4_08 by the same procedure.

PROBLEM 1 - poor image contrast with cine phase contrast scan
previously addressed in version D patch_90_M4_08

CQA - 1019049 CNV3-poor imaging w/Cine Phase Contrast scans
SPR - MRIge69155 CNV3-poor imaging w/Cine Phase Contrast scans
SPR - MRIge67315 fastcardPC full echo is offset
SPR - MRIge68880 Flow Velocity Annotation Carry-over From Prior RX

Description -

Fastcard Phase Contrast images in Release 9.0 are not the same as in Release 8.4. A User CV (opuser2) was implemented to allow users to use the magnitude weighting mask or not. The default value for opuser2 was correctly only when Fastcard PC is started independently. If the user selects the sequence from a previously prescribed FGRE-based sequence, such as through copy/paste/view-edit, the default becomes a zero.

Workaround -

The user can avoid this problem by always checking opuser2.

The fix is in the following subsystems:

PSD: fgre

PROBLEM 2 - double obliques in fiesta2d

CQA - 10112877 LEO No double obliques
SPR - MRIge71733 LEO No double obliques

Description -

fiesta2d won't allow double obliques to be scanned. It asks for an increase in the FOV that is never able to be satisfied, even when 48cm is entered. Single oblique is ok. This problem stems from a hard-coded max FOV limit, that is opened up in this patch. The problem never existed in the CNV3 or CNV4 versions of fiesta2d.

Workaround -

None

The fix is in the following subsystems:

PSD: fgre

Patch_90_M4_09 Rev D

This patch contains fixes for four problems found with the 9.0 release, none of which have been previously addressed. The size of this download for this patch is approximately 1.8 megabytes, compressed.

PROBLEM - Elliptic Centric Ordering unavailable in efgre3d without a SmartPrep key

SPR - MRIge71565 Elliptic Centric ordering should be available w/o a key

Description -

The Elliptic Centric ordering option was available for all Fast 3D TOF GRE and SPGR scans since the 8.3 release. However, after several changes in 9.0, the option became enclosed inside code that was only active with the SmartPrep key. This patch restores the 8.3 functionality. The new ordering options (centric, reverse elliptical centric, and reverse centric) are only available for Fast 3D TOF GRE and SPGR if the SmartPrep key is present. However, Elliptical Centric is available without the key.

The fix is in the following subsystems:

PSD: efgre3d

Patch_90_M4_11 Rev G

This patch contains fixes for two problems found with the 90 'LEO1' release, none of which have been previously addressed. The size of the download for this patch is approximately 1.2 megabytes, compressed.

PROBLEM 1 - No Fat Sat with SSFSE if using Manual Prescan

SPR - MRIge69015 FATSAT doesn't function on 1st scan with ROCKER & SSFSE

Description -

Fat Saturation does not work when running SSFSE with Manual Prescan, even though the annotation shows that it did.

Workaround -

Use Auto Prescan

The fix is in the following subsystems:

PSD: ssfse

PROBLEM 2 - Fractional NEX problems in SSFSE

SPR - MRIge71989 Fractional NEX reconstruction does not work properly in SSFSE

Description -

When Fractional NEX Optimization was ON, blurring images, elongated organs and incorrect object positions were seen. This was seen with fractional NEX reconstruction code only when arbitrary values such as 0.83 NEX were used. SSFSE uses 0.83 NEX.

Workaround -

None.

The fix is in the following subsystems:

PSD: ssfse

Patch_90_M4_12 Rev K

PROBLEM 1 - 1RR/2RR slices difference in fse
previously addressed in version G of this patch

SPR - MRIge69226 Aarhus -- with RTr large diff for # slices between 1 RR & 2RR

Description -

There is a large difference between number of slices per respiratory interval when using respiratory triggering.

The calculations for seqtype = TYPRTG includes a multiplication by the number of respiratory cycles in both imgtimutil and in maxslquant. The maximum slice quantity then goes up as the number of respiratory cycles squared. For cardiac

calculations, the available image time is not multiplied by the number of cardiac cycles, but for type TYPXRR, the number of slices is.

Workaround -
None.

The fix is in the following subsystems:
PSD: fse

PROBLEM 2 - 1RR/2RR slices difference in fse (fseopt and fse-xl)

SPR - MRlge69226 Aarhus -- with RTr large diff for # slices between 1 RR & 2RR

Description -
The fix for problem covered only the fse pulse sequence, now the fix is expanded to cover fseopt and fse-xl too.

Workaround -
None.

The fix is in the following subsystems:
PSD: fseopt, fse-xl

Patch_90_M4_13 Rev J

PROBLEM 1 - EPI ghosting on Twin
previously addressed in version H patch_90_M4_13

SPR - MRlge69516 WHOLE/ZOOM file name selection

Description -
Added code to read in delay.dat.ZOOM & delay.dat.WHOLE based on gradient coil select by the user. This change is needed so the proper delays are used to set up the readout window position relative to the read gradient position. If the read window and read gradient are misaligned, high-frequency ghosting results on ramp sampled EPI.

Workaround -
None.

The fix is in the following subsystems:
PSD: epi, epi2, and epiVP
TOOLS: grafidy and ecmt coefficient files

Note that in order to fully implement this fix, a Field Engineer needs to perform certain operations at the site, after installing this patch. See the service instructions that accompany CD 2327548-3.

PROBLEM 2 - Remove "DOT" artifact in DWEPI

SPR - MRlge71771 rhblne NOT set correctly in epi2.e

Description -
This problem pertains to multi-nex DW-EPI. The problem is that for 2 NEX scans, we normally do not collect baseline data from the fast-receiver to correct for DC artifacts. However, with DW-EPI for multi-NEX scans the data are averaged in image space. Thus any DC artifact does not subtract out. Baselines are acquired in epi2 but only at specific combinations of NEXT. The solution is to

acquire baselines for ALL NEX when the fast receiver is used.

Workaround -
None.

The fix is in the following subsystems:
PSD: epi2

Patch_90_M4_14 Rev L

FEATURE - high order shim

This feature was previously packaged in preliminary form in patch_90_M4_03, and later released in patch_90_M4_10, and in version "i" of this patch.

Description -

The high order shim option delivers a resistive shim supply to connect to the 2nd order shim coils already existing in the Twin Resonator Module (TRM). This option allows the operator to use localized shimming in-vivo on a per patient basis by means of a special shim series that is used as often as necessary during the Exam.

The feature is implemented in the following subsystems:

HOST: miscellaneous prescan tools and scripts

PSD: sprl_hos

HOST: scn executable

HOST: error message database

HOST: protocols and config files

REALTIME: tardis.o

PROBLEM 1 - fiesta2d not available without fiesta3d option key

This problem was previously addressed in patch_90_M4_01, patch_90_M4_03, patch_90_M4_10, and version "i" of this patch.

SPR - MRIGe70815 fiesta2d psd selection is not available

Description -

When both fiesta2d and fiesta3d option keys are installed, both fiesta2d and fiesta3d can be selected with the Fiesta Imaging Option. When only the fiesta3d option key is installed, only fiesta3d can be selected with the Fiesta Imaging Option. However, when only the fiesta2d option key is installed, the Fiesta Imaging Option is unavailable.

The fix is in the following subsystems:

HOST: scn executable

PROBLEM 2 - 3 plane graphic rx may show localizers from prior exam

previously addressed in patch_90_M4_02, and version "i" of this patch.

SPR - MRIGe70890 Stanford:called wrong pt into 3planeGRX

Description -

Under some circumstances, 3 plane graphic rx may default to using localizer images from prior exam and patient.

Workaround -

Use list-select to choose a series from the current exam. After that, 3 plane graphic rx will take images from the current exam.

The fix is in the following subsystems:
HOST: 3 plane graphic rx shared library

PROBLEM 3 - Smart Prep Tracker in wrong place
previously addressed in patch_90_M4_02, and version "i" of this patch.

CQA - 10112117 LEO 3 plane GRX fails
SPR - MRIge71418 LEO 3 plane GRX fails
CQA - 10112443 LEO SmartPrep, no detection
SPR - MRIge71549 LEO SmartPrep, no detection
CQA - 1020054 LEO rotate tracker, corrupt blood flow
SPR - MRIge71850 LEO rotate tracker, corrupt blood flow

Description -

Using contrast detection with 3 plane graphic: if tracker is placed on one axis and scan is prescribed off different, tracked either shifts location or just doesn't work.
NO DETECTION WITH SMARTPREP OR DETECTION TOO LATE WITH NO PATIENT MOTION AND CORRECT TRACKER PLACEMENT. PROBLEM OCCURRED FOR CAROTIDS SCAN WITH NARRAY COIL
Customer has problem with Tracker since 9.0 upgrade. If they rotate Tracker then they get corrupt blood flow. If they scan without rotating Tracker, everything is fine. This changed with 9.0.

Workaround -
None.

The fix is in the following subsystems:
HOST: 3 plane graphic rx shared libraries
HOST: Graphic Rx server executable

PROBLEM 4 - spectro oblique rotations are incorrect
previously addressed in version "i" of this patch

SPR - MRIge71838 Spectroscopy: Oblique prescription rotations incorrect

Description -

Workaround -

The fix is in the following subsystems:
PSD: presscsi, prose, steamcsi
HOST: scn executable
REALTIME: tardis.o

PROBLEM 5 - phase/freq direction issue with poor MRCP IQ
previously addressed in version "i" of this patch

SPR - none

Description -

The code which decided which logical direction is Phase and which is Frequency was not completely general for double and triple obliques, which are easier to accomplish with 3-Plane Graphic Rx than before. This Frequency/Phase direction logic was enhanced to better handle these cases. The fix also helps with consistency of Frequency/Phase directions for multi-group multi-angle prescriptions that are typically used in MRCP prescriptions.

Workaround -

None.

The fix is in the following subsystems:

HOST: scn executable

PROBLEM 6 - film, archive, network resume after RTIA

SPR - MRlge72506 RTIA should read ENV to determin if IOS process dhoulds be pau

Description -

Current design of RTIA when active will pause Filming, Archive and Network processes and resume the processes when RTIA session end. Intermittent IOS jobs not being resume properly has been reported since ASP2. Customer are not very happy with this intermittent issue. (CSO: 0105979-ASP2 0200730-CNV4)

Workaround -

Manually resume jobs.

The fix is in the following subsystems:

HOST: scn executable

PROBLEM 7 - PROSE prescan fix for multi-coil

SPR - MRlge72657 Spectro Prescan: Multi-coil problem

Description -

While using the Spectro Prescan page on a PROSE exam, it was noticed that the signal did not appear to represent the coil selected on the Parameters page. Further investigation showed that when all four coils of the four coil array were active in the selected CoilConfig entry, choosing between the coils on both the SpectroPrescan and Manual Prescan page worked as expected.

Workaround -

None.

The fix is in the following subsystems:

REALTIME: CERD

REALTIME: tardis.o

PROBLEM 8 - PROSE shifted lipid peak

SPR - MRlge73052 PROSE: over-volume Rx factors need RAS designation

Description -

Prostate CSI acquisition displays too much susceptibility shifted lipid signal. One cause appears to be the over-prescription factor in the A/P direction. The default over-prescription factor (increase in the excited voxel size versus final size as limited by saturation bands) is 1.3 for all three RF pulses.

Workaround -

None.

The fix is in the following subsystems:

PSD: presscsi, prose, steamcsi

PROBLEM 9 - PROSE protocol fix

SPR - MRlge73416 PROSE: Protocol updates needed

Description -

The GE protocols for the Prostate imaging + PROSE spectroscopy product have the Frequency and Phase resolutions switched. The larger resolution for these protocols should be in the R/L direction. For the coils used in this protocol, this is the frequency direction.

The protocol entries for MATRIXX and MATRIXY need to be swapped such that MATRIXX = 16 and MATRIXY = 8.

Workaround -

None.

The fix is in the following subsystems:

HOST: GE Protocols

PROBLEM 10 - GrxServer memory leak

CQA - 1020148 LX PSD Fails to start errors

SPR - MRlge71891 LX PSD Fails to start errors

CQA - 1022120 9.0 sluggish or slow performance

SPR - MRlge72951 9.0 sluggish or slow performance

CQA - 1022126 9.0 sluggish performance

SPR - MRlge72952 9.0 sluggish performance

SPR - MRlge71410 Thomas Jefferson scan exit due to oversized grxServer

Description -

System gradually slows down and becomes sluggish after use of the 3-plane graphic feature. The problem is due to a programming error where memory is allocated and never freed, resulting a process that grows in memory size over time. Symptoms include sluggish response, and eventually intermittent "psd fails to start" error messages

Messages appear in the SYSLOG, such as:

Nov 29 13:16:14 3B:MRSD_OC0 syslog: grxServer: CUP ERROR in file:

cupMakeCore.line: 270

Nov 29 13:16:14 3B:MRSD_OC0 syslog: program grxServer: pid 15455: signal 11:

fe core.grxServer.29Nov131553

Nov 29 13:16:28 1A:MRSD_OC0 unix: ALERT: scn [14810] - out of logical swap spa during fork/sproc - see swap(1M)

Nov 29 13:18:37 3D:MRSD_OC0 ppp[271]: olc-pm3 LCP1: receive

Terminate-Request05 01 00 04

Workaround -

Reboot Signa daily.

The fix is in the following subsystems:

HOST: grxServer executable

PROBLEM 11 - Images Under Different Patient Name

FPR - 10110784 IMAGES UNDER DIFFERENT PATIENT

SPR - MRlge71614 IMAGES UNDER DIFFERENT PATIENT

CQA - 1021700 9.0 His/Ris, PID, Mitra

SPR - MRlge72620 9.0 His/Ris, PID, Mitra

Description -

Patient images were found in the exam of a previously scanned patient. The patient images were mislabeled with this earlier scanned patient's name, ID number, and all other personal information. The patient's file that the images were placed into was from a patient scanned 12 hours earlier.

Workaround -
None.

The fix is in the following subsystems:
HOST: scn executable

PROBLEM 12 - PROSE recon line broadening

SPR - MRIge73642 MRS: CSI recon apodization improvement

Description -
The fixed apodization in the CSI reconstruction package for both the Brain but mostly the Prostate package was found to be insufficient and less than the ideal as established by the UCSF evaluation team.

Workaround -
None.

The fix is in the following subsystems:
PSD: presscsi, prose, steamcsi
REALTIME: tardis.o

PROBLEM 13 - Tracker rx lost in smart step

CQA - 1022844 9.0 SmartPrep bolus detect
SPR - MRIge73434 9.0 SmartPrep bolus detect
CQA - 1024268 9.0 Smartprep - Gradients inactive
SPR - MRIge74481 9.0 Smartprep - Gradients inactive
CQA - 1022846 9.0 SmartPrep bolus detect
SPR - MRIge73437 9.0 SmartPrep bolus detect

Description -
When the user prescribes the tracker prescription, switches back to GraphicRx, and accepts the Graphic Rx while the UI is coming up, data gets erased, because X-windows gets a new event of higher priority. The flag which represents the Tracker's presence, gets set initially, when the user prescribes the tracker, and gets reset when the user does erase of the tracker. But in this case, this flag is not reset, causing the Save Series button to get enabled. Thus the user is able to save the series without even having the tracker prescription, and thus the rotation matrixes are wrong.

Since this is a timing issue, we now set the Busy Cursor when the user switches from the Tracker screen to GraphicRx screen. To be safe, we also check for the Tracker prescription whenever the user presses the accept of the Graphic Rx screen.

Workaround -
The user can wait for graphic rx to finish before doing Save Series.

The fix is in the following subsystems:
HOST: 3 plane graphic rx shared libraries

PROBLEM 14 - EPI is not annotated on the images

SPR - MRIge74560 Bay4 - EPI is not annotated on the images

Description -

"EPI" is missing from pulse sequence identification in annotation

Workaround -
none

The fix is in the following subsystems:
HOST: scn executable

Patch_90_M4_15 Rev J

PROBLEM 1 - Early bolus detection in SmartPrep

SPR - MRIge73007 SmartPrep SPU promote for minizing bolus early detection

Description -
Update the SmartPrep SPU algorithm to minimize early detection of bolus as experienced with 9.0 M4 release.

Workaround -
None.

The fix is in the following subsystems:
REALTIME: SPU

Patch_90_M4_17 Rev J

This patch contains the fix for one problem found with the 90 'LEO1' release, which has not been addressed previously. The size of this download for this patch is approximately 1.3 megabytes, compressed.

PROBLEM - 3DFSE aborts on second slab

CQA - 1022377 9.0 3DFSE 2-slab needed
SPR - MRIge73143 9.0 3DFSE 2-slab needed
CQA - 1023388 9.0 3D FSE- Scan Aborts
SPR - MRIge73767 9.0 3D FSE- Scan Aborts
CQA - 10112005 LX / 3DFSE Recons only one slab
SPR - MRIge71366 LX / 3DFSE Recons only one slab

Description -
When doing a 3D FSE prescription with 2 slabs, the first series scans correctly, but then the scan aborts and only that slab is reocnstructed. The error logged is "2226578 PSD Programming Error: The PSD was still preparing a sequence when an EOS Interrupt occurred. Lengthen TR or reduce the number of slices."

Workaround -
If you only perscibe one slab, it works fine. But the workarounds recommended by the error message result in unacceptable image quality.

The fix is in the following subsystems:
PSD: 3dfse